

Lecture 24

Eigenvalue III: Diagonalization

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Review and Outline

3 mins

Today's Lecture: Outline and Learning Goals

Main topic: Properties

1. Similar Matrix

2. Diagonalization

3. Application: Dynamics

After the lecture, you should be able to

1. Tell concepts of similar matrices, and property of them

2. Describe the concept and conditions for diagonalizable

3. Write the equations of dynamics & analyze the dynamics

Eigenvalues and Eigenvectors

Definition 21.1 (Eigenvalues and Eigenvectors)

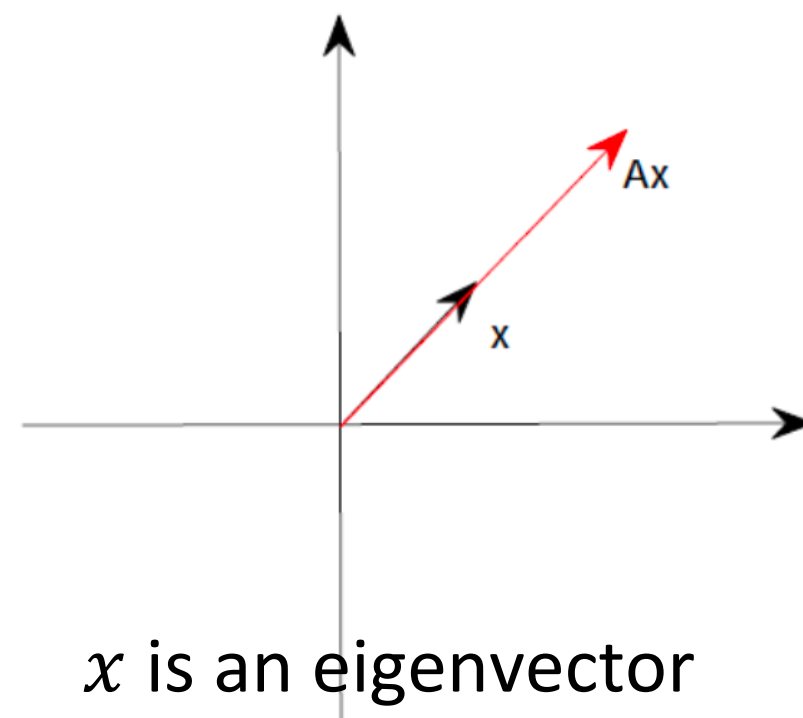
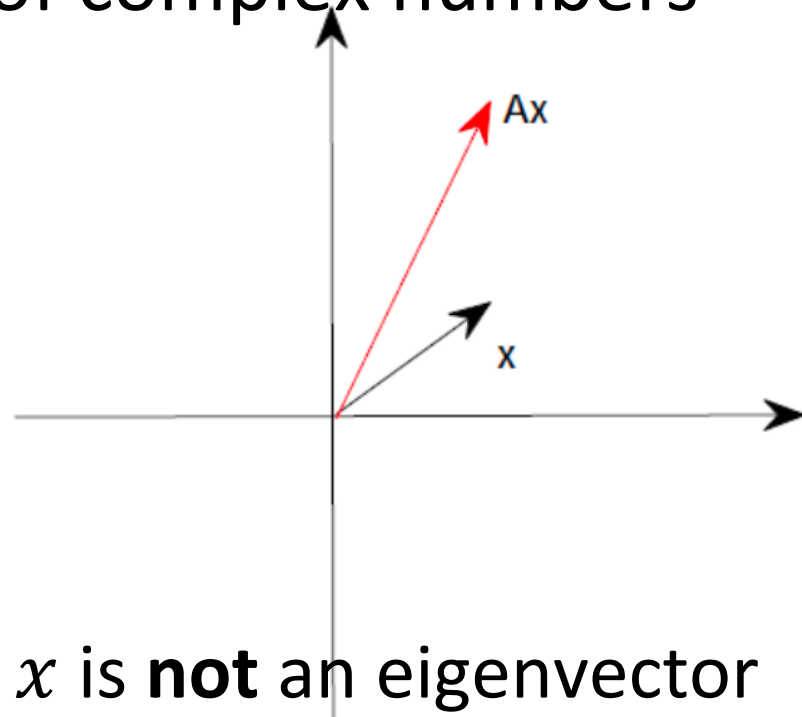
Let $A \in \mathbb{C}^{n \times n}$ be a square matrix.

If there exists a scalar λ ($\in \mathbb{R}$ or $\mathbb{C}$) and a **nonzero** vector x such that

$$Ax = \lambda x,$$

then λ is called a (real or complex) **eigenvalue** and x is called an **eigenvector** with respect to (or associated with; corresponding to) λ .

$\mathbb{C} :=$ set of complex numbers



Procedure to Find Eigenvalues/Eigenvectors

Step 1: Solve $\det(A - \lambda I) = 0$ and get n roots $\lambda_1, \dots, \lambda_n$

Solving single-var polynomial equation

Step 2: For each λ_i , find the eigenspace $N(\lambda_i I - A)$

Solving a number of (up to n) linear systems of equations.

Any nonzero vector in $N(\lambda_i I - A)$ is an eigenvector corresponding to λ_i .

$$N(\lambda_i I - A) = \left\{ \text{eigenvectors of } A \text{ with corresponding to the eigenvalue } \lambda_i \right\} \cup \{0\}$$

Fact:

Any $n \times n$ (no matter real or complex) matrix A has n **complex eigenvalues** (counting multiplicity).

Characteristic polynomial $p_\lambda(A) = \det(A - \lambda I)$ has n roots.

Short Summary of Results

Suppose $A \in \mathbb{C}^{n \times n}$.

$D = \text{diag}(\lambda_1, \dots, \lambda_n)$, $V = [v_1, \dots, v_n]$, **where** $v_j \in \mathbb{C}^n$, $\lambda_j \in \mathbb{C}$.

$$Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n,$$

$$\iff AV = VD, \quad (*)$$

Suppose $A \in \mathbb{R}^{n \times n}$.

$D = \text{diag}(\lambda_1, \dots, \lambda_n)$, $V = [v_1, \dots, v_n]$, **where** $v_j \in \mathbb{C}^n$, $\lambda_j \in \mathbb{C}$.

$$Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n,$$

and $\{v_1, \dots, v_n\}$ form an orthonormal basis of $\mathbb{R}^n$
(*imply $v_j \in \mathbb{R}^n$, then imply $\lambda_j \in \mathbb{R}$*).

$$\iff A = VDV^T, \text{ where } V \text{ is a real orthogonal matrix, } D \text{ is a real diagonal matrix}$$

$$\iff A \text{ is real symmetric.}$$

Formal version of Spectral Theorem

Theorem 23.1 [Spectral Theorem]

Any real symmetric matrix A can be written as

$$A = VDV^T = \sum_{j=1}^n \lambda_j \mathbf{v}_j \mathbf{v}_j^T. \quad (*)$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a real diagonal matrix,
 $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ is a real orthogonal matrix.

(*): Eigenvalue decomposition (EVD) or eigendecomposition of A .

Property 23.1: All eigenvalues are real.

Property 23.2: Eigenvectors can form an orthonormal basis of $\mathbb{R}^n$.

(23.2a) Eigenvectors can form a basis of $\mathbb{R}^n$.

(23.2b) Can pick an eigenbasis that is orthonormal set.

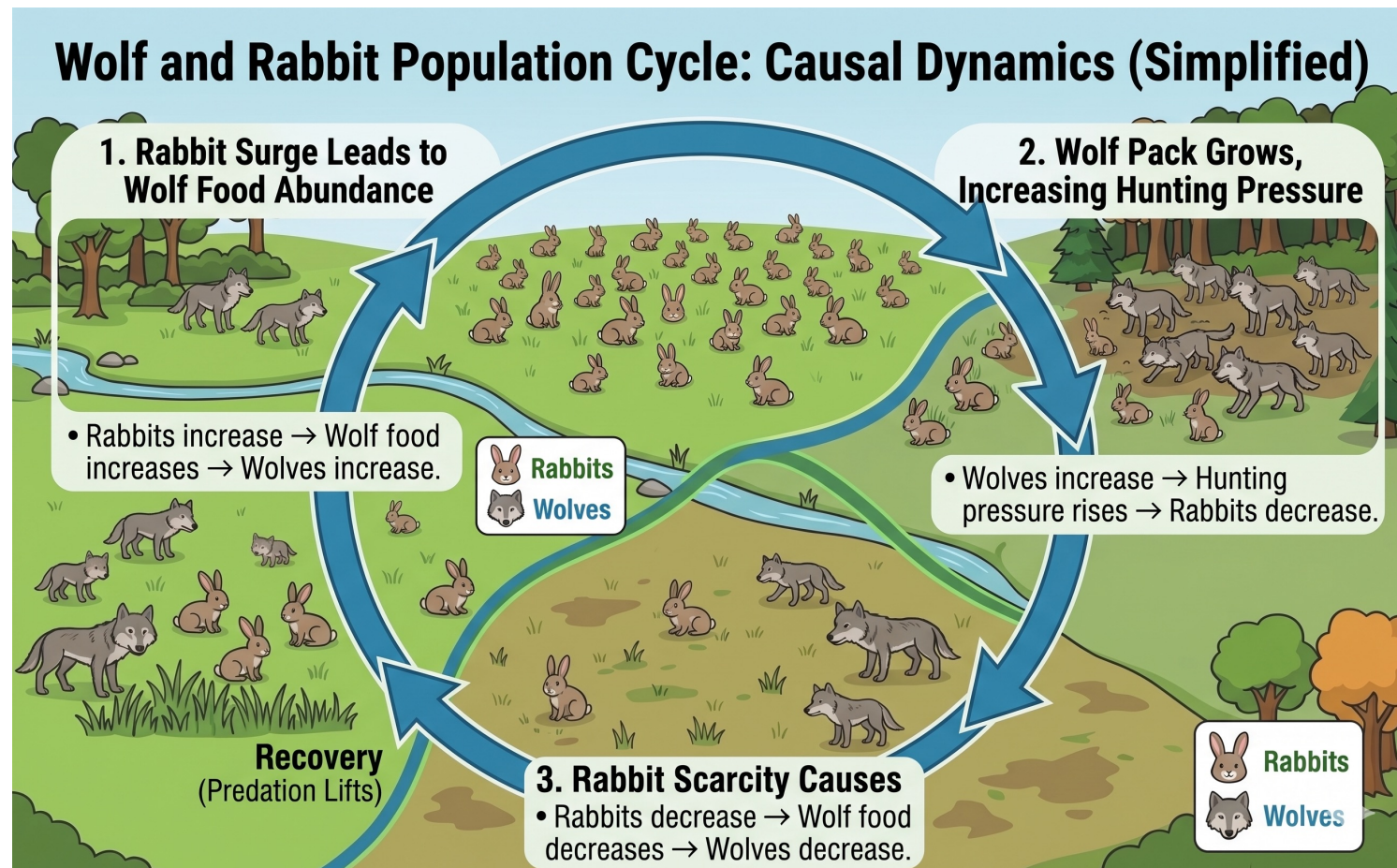
Part I Motivation: Evolution Dynamics

15 mins

Dynamics of Wolves and Rabbits

Consider a prairie with wolves and rabbits.

- Rabbits increase $\rightarrow$ Wolf food $\rightarrow$ Wolves $\rightarrow$
- Wolves increase $\rightarrow$ Hunting pressure rises $\rightarrow$ Rabbits $\rightarrow$
- Rabbits decrease $\rightarrow$ Wolf food decreases $\rightarrow$ Wolves $\rightarrow$



The Unanswered Question:
An endless cycle?

- Simple intuition is NOT enough.
- *Will they oscillate back and forth forever?*
- Will one species accidentally drive the other to extinction?

Need quantitative analysis! (定量分析)

Predator-Prey Model (捕食者-猎物模型)

Dynamics of wolves and rabbits.

Type Number in year t

Wolves y_t

Rabbits x_t

State vector: $z_t = \begin{bmatrix} x_t \\ y_t \end{bmatrix}$

Predator (Wolves)

- **Predation gain rate:** $\beta_1 = 0.3$
→ Each rabbit supports the survival/growth of 0.3 wolves per year.
- **Natural survival rate:** $\beta_2 = 0.5$
→ 50 % of wolves survive each year without rabbits.

of wolves in new year =
 $0.3 * (\# \text{ of rabbits}) + 0.5 * (\# \text{ of wolves})$

$$y_{t+1} = 0.3 x_t + 0.5 y_t$$

Prey (Rabbits)

- **Natural growth rate:** $\alpha_1 = 1.2$
→ Rabbits grow by 20 % annually without predation.
- **Predation impact:** $\alpha_2 = -0.2$
→ Each wolf reduces the rabbit population by 0.2 per year.

of rabbits in new year =
 $1.2 * (\# \text{ of rabbits}) - 0.2 * (\# \text{ of wolves})$

$$x_{t+1} = 1.2 x_t - 0.2 y_t$$

Dynamics

(∞ -time transformation)

Two sequences, $\{u_k\}$ and $\{v_k\}$, where $k=0,1,2,\dots$

The sequences evolve according to these **equations**:

$$\begin{cases} u_{k+1} = \alpha_1 u_k + \alpha_2 v_k, \\ v_{k+1} = \beta_1 u_k + \beta_2 v_k, \end{cases}$$

- $\alpha_1, \alpha_2, \beta_1, \beta_2$ are constant coefficients (real numbers).
- These coefficients determine how u_k and v_k influence each other's future values.

Matrix form:

$$\underbrace{\begin{pmatrix} u_{k+1} \\ v_{k+1} \end{pmatrix}}_{\vec{z}_{k+1}} = \underbrace{\begin{pmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} u_k \\ v_k \end{pmatrix}}_{\vec{z}_k},$$

$$\vec{z}_{k+1} = A \vec{z}_k, \quad k=0,1,2,\dots$$

where $\vec{z}_k = \begin{pmatrix} u_k \\ v_k \end{pmatrix}$ is the **state vector** at step k .

Long-term Behavior: Computing A^k

$$\vec{z}_1 = A \vec{z}_0$$

$$\vec{z}_2 = A(A \vec{z}_0) = A^2 \vec{z}_0$$

$$\vec{z}_3 = A(A^2 \vec{z}_0) = A^3 \vec{z}_0$$

$$\underline{\vec{z}_k} = \underline{A^k} \underline{\vec{z}_0}$$

Predator-Prey Model (捕食者-猎物模型)

Write the update equations in matrix form:

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} 1.2 & -0.2 \\ 0.3 & 0.5 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix}$$

$$z_{t+1} = Az_t$$

Q1: Compute # of wolves and rabbits in year t.

Q2: Will # of wolves/rabbits grow to ∞ or diminish to 0?

Next:

Compute the evolution of numbers, for 1,2,3,4,5,

Computing Numbers Naively

Assume initial population: $z_0 = [100, 40]^T$.

Evolution Of V_k (Rounded) For $K=0..5$

	k	x_k	y_k
1	0	100.0	40.0
2	1	112.0	50.0
3	2	124.4	58.6
4	3	137.56	66.62
5	4	151.748	74.578

$A z_0$

$A^2 z_0$

$A^3 z_0$

$A^4 z_0$

Another method

When A is $10^6 \times 10^6$
 $10^9 \times 10^9$

A is hard to compute.

Question: Is there a simpler formula to compute z_t ?

Other Examples of Dynamics

Epidemic Waves (SARS / Flu / COVID)

Three types:

Susceptible (Healthy), Infected, Recovered

Numbers of each type of people:
depend on other types of people.

Can build evolution equations.

Part II Similar Matrices

17 mins

Similar Matrices

Definition 24.1 (Similar matrix 相似矩阵)

If two $n \times n$ matrices A and B satisfy $A = SBS^{-1}$ for some $S \in \mathbb{R}^{n \times n}$, then we say A and B are similar (相似), denoted as $A \sim B$.

e.g. 2×2 matrices $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ are similar, since $A = SBS^{-1}$ with

$$SBS^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$$

NB: If a matrix is similar to identity matrix, then it is an identity matrix as well. $S \cdot I \cdot S^{-1} = S \cdot S^{-1} = I$

in general, $A = S \cdot B \cdot S^{-1} \neq S \cdot S^{-1} \cdot B = B$.

Property of Similar Matrices

Claim 24.1

Suppose $A, B \in \mathbb{R}^{n \times n}$ and are similar, i.e., there exists an invertible matrix S such that $A = SBS^{-1}$.

If λ is an eigenvalue of A , then λ is an eigenvalue of B .

Proof:

$$Ax = \lambda x \Rightarrow SBS^{-1}x = \lambda x$$

$$\Rightarrow \cancel{S^{-1}} \cancel{S} B S^{-1} x = \cancel{S^{-1}} \cdot \underline{\lambda} x$$

$$\Rightarrow B(S^{-1}x) = \lambda(S^{-1}x)$$

$$\Rightarrow By = \lambda y \quad (\text{let } y = S^{-1}x)$$

$$(y = S^{-1}x \neq 0, \text{ since } S \text{ invertible and } x \neq 0)$$

$$(If \ S^{-1}x = 0 \Rightarrow x = S \cdot 0 = 0)$$

$$\begin{aligned} SBS^{-1}x &= \lambda x \\ \Rightarrow B S^{-1}x &= S^{-1}(\lambda x) \end{aligned}$$

Does This Imply "Same Eigenvalues"?

$$A = S \cdot B \cdot S^{-1}$$

C1 If λ is an eigenvalue of A , then λ is an eigenvalue of B .

Similarly, we can prove:

$$S^{-1} \cdot A \cdot S = B.$$

C2 If λ is an eigenvalue of B , then λ is an eigenvalue of A .

Question:

Do C1 & C2 imply "A and B have the same eigenvalues"?

No (under interpretation of multiset)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\text{EIG}(A) = \# \{1, 1, 2\}$$

$$\text{EIG}(B) = \# \{1, 2, 2\}.$$

Product of Eigenvalues

Proposition 24.1 (Similar matrices have same eigenvalues)

Suppose $A, B \in \mathbb{R}^{n \times n}$ are similar, then $\text{EIG}(A) = \text{EIG}(B)$.

Recall : $\text{EIG}(A)$ is the **multiset** of eigenvalues of matrix A .

[Write as matrix product]

Comment: The proof of Claim. 24.1 is not enough for Prop. 24.1, since it does not consider multiplicity (explained in Lec 24).

[Property of matrix product]

Proof: Denote $p_A(\lambda) = \det(\lambda I - A)$, $p_B(\lambda) = \det(\lambda I - B)$,

Assume $B = S^{-1}AS$, then

$$\begin{aligned} p_B(\lambda) &= \det(\lambda I - B) \\ &= \det(S^{-1}(\lambda I - A)S) \\ &= \det(S^{-1}) \det(\lambda I - A) \det(S) \\ &= \det(\lambda I - A) = p_A(\lambda) \end{aligned}$$

$$\begin{aligned} \lambda I - B &= \lambda I - S^{-1}AS \\ S^{-1}(\lambda I - A)S &= S^{-1}\lambda I S - S^{-1}AS \\ &= \lambda S^{-1}S - S^{-1}AS \\ &= \lambda I - S^{-1}AS \end{aligned}$$

$\det(S^{-1})\det(S)=1$

The eigenvalues of A and B are the roots of $p_A(\lambda)$ and $p_B(\lambda)$ respectively.

$\therefore A$ and B have the same eigenvalues. $\square$

$$\begin{aligned} \lambda I - B &= S^{-1}\lambda I S - S^{-1}AS \\ &= S^{-1}(\lambda I - A)S \end{aligned}$$

Useful Tool

Claim: $\det(XYZ) = \det(XZY)$.

Proof: $\det(XYZ) = \det(X) \det(Y) \det(Z)$
 $= \det(X) \det(Z) \det(Y)$
 $= \det(XZY).$

$$XYZ \neq XZY.$$

Proof of Eigenvalues of Similar Matrices

Think: What's in your toolbox for eigenvalues?

$$Ax = \lambda x \text{ (Claim 24.1)}$$

Tool 1: Definition.

λ is a root of $\det(\lambda I - A)$

Tool 2: Characteristic polynomial. (see Prop. 24.1)

Tool 3: Eigenvalue decomposition (for symmetric matrices.)

Part III Diagonalizable and Computing A^k

20 mins

Matrix Form of Eigen-equations

Recall: Matrix Form of Eigen-equations.

Lemma 23.1 & 23.2 [eigen-equations in matrix form]

Suppose A is a square matrix.

$Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n$, where $v_j \in \mathbb{C}^n, \lambda_j \in \mathbb{C}$.

$$\iff AV = VD,$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ diagonal, $V = [v_1, \dots, v_n]$.

Last time:

If V is an orthogonal matrix, we can rewrite as $A = \text{sth.}$

If V is NOT necessarily an orthogonal matrix:

Can we further rewrite as $A = \text{sth.}$?

$$AV = VD \Rightarrow A = \underbrace{VD \cdot V^{-1}}_{\substack{\downarrow \\ \text{diagonal}}} \Rightarrow A \sim \underbrace{D}_{\substack{\downarrow \\ \text{diagonal}}}$$

$V^{-1} \exists$

Matrix Form of Eigen-equations

Definition 24.2 (Diagonalizable 可对角化)

If a square matrix A is similar to a diagonal matrix, i.e., $A = SDS^{-1}$ for some diagonal matrix D , then we say A is diagonalizable.

E.g.

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}_S \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}}_D \underbrace{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}}_{S^{-1}}$$

$$A \cdot S = SD \implies A = SDS^{-1} \text{ if } S^{-1} \exists$$

Diagonalizable

Theorem 25.1 [Suffi. & Necc. Conditions for Diagonalizable: n indep. eigenvectors]

An $n \times n$ matrix A is diagonalizable $\Leftrightarrow$ eigenvectors of A can form a basis.

i.e. eigenbasis of $A \exists$.

Proof:

A is diagonalizable

$\Leftrightarrow A = S \cdot D \cdot S^{-1}$ for some $S = [v_1, \dots, v_n] \in \mathbb{C}^{n \times n}$ and

definition $D = \text{diag}(\lambda_1, \dots, \lambda_n) \in \mathbb{C}^{n \times n}$

$\Leftrightarrow AS = SD$, for some invertible $S = [v_1, \dots, v_n] \in \mathbb{C}^{n \times n}$
multiply and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$

$\Leftrightarrow Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n$, for some $\{v_j, j = 1, \dots, n\}$
Vector form forming a basis of $\mathbb{C}^n$, and some $\lambda_j \in \mathbb{C}, j = 1, \dots, n$

$\Leftrightarrow A$ has n eigenvectors $v_1, \dots, v_n$ that form a basis of $\mathbb{C}^n$.

Not All Matrices Are Diagonalizable

Recall (Lec 23): The eigenvectors of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ do NOT form a basis of $\mathbb{C}^2$.

No eigenbasis for $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
So, $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is NOT diagonalizable.

Fact: Not all square matrices are diagonalizable.

Another quicker way to check:

Eigenvalues of $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ are 1,1.

Assume $A = SDS^{-1}$, then $\text{EIG}(D) = \text{EIG}(A) = \{1, 1\}$.

A diagonal matrix D has eigenvalues 1,1, thus $D = \text{diag}\{1, 1\} = I_2$.

Thus $A = S I_2 S^{-1} = I_2$, contradiction.

Sufficient Condition for Diagonalizable

Proposition 25.2 (distinct eigenvalues $\rightarrow$ diagonalizable)

Suppose an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable.

Lemma 1: Given a matrix A , if $\lambda_1, \lambda_2, \dots, \lambda_k$ are distinct eigenvalues of A with corresponding eigen vectors $x_1, x_2, \dots, x_k$, then the **vectors $x_1, x_2, \dots, x_k$ are linearly independent.**

Skip the proof here.

last time. symmetric matrix, different eig λ_1, λ_2

$\Rightarrow$ orthogonal vec v_1, v_2

this time: matrix, different eig. λ_1, λ_2

$\Rightarrow$ indep v_1, v_2

Sufficient Condition for Diagonalizable

Proposition 25.2 (distinct eigenvalues $\rightarrow$ diagonalizable)

Suppose an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable.

Lemma 1: Given a matrix A , if $\lambda_1, \lambda_2, \dots, \lambda_k$ are distinct eigenvalues of A with corresponding eigen vectors $x_1, x_2, \dots, x_k$, then the vectors $x_1, x_2, \dots, x_k$ are linearly independent.

Proof of Prop. 25.2 based on Lemma 1:

By this claim, and condition of Prop. 25.2,

we can find n linearly independent eigenvectors in $\mathbb{C}^n$.

They form a basis of $\mathbb{C}^n$. (eigenbasis)

By Thm 25.1, A is diagonalizable.

Diagonalizable Makes A^k Easy to Compute

Claim: If $A = SDS^{-1}$ for some diagonal matrix D , then $A^k = SD^kS^{-1}$.

In order to guess the result, we try a simple case:

$$A^2 = (SDS^{-1})(SDS^{-1}) = SD(\cancel{S^{-1}S})D S^{-1} = SD^2S^{-1}.$$

Guess: $A^k = SD^kS^{-1}$

Prove by induction (归纳法)

Assume $A^{k-1} = SD^{k-1}S^{-1}$

Then $A^k = A^{k-1} \cdot A = (SD^{k-1}\cancel{S^{-1}})(\cancel{S}DS^{-1}) = SD^kS^{-1}.$

Exercise: Judgement

Judgement: A real square matrix is always similar to a real diagonal matrix.

F.

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Judgement: If $A, B \in \mathbb{R}^{n \times n}$ are similar, then A and B have the same multiset of eigenvalues.

T.

Judgement: If $A, B \in \mathbb{R}^{n \times n}$ have the same multiset of eigenvalues, then they are similar.

F

$$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad 1, 1,$$

Judgement: If $A \in \mathbb{R}^{n \times n}$ has two eigenvalues that are different, then A is diagonalizable.

F

no guarantee.
n distinct: guarantee diagonalizable

Part IV Application of Diagonalization: Dynamics

15 mins

Dynamics

Two sequences, $\{u_k\}$ and $\{v_k\}$, where $k=0,1,2,\dots$

The sequences evolve according to these **equations**:

$$\begin{cases} u_{k+1} = \alpha_1 u_k + \alpha_2 v_k, \\ v_{k+1} = \beta_1 u_k + \beta_2 v_k, \end{cases}$$

- $\alpha_1, \alpha_2, \beta_1, \beta_2$ are constant coefficients (real numbers).
- These coefficients determine how u_k and v_k influence each other's future values.

Matrix form:

$$\underbrace{\begin{pmatrix} u_{k+1} \\ v_{k+1} \end{pmatrix}}_{\vec{z}_{k+1}} = \underbrace{\begin{pmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} u_k \\ v_k \end{pmatrix}}_{\vec{z}_k},$$

where $\vec{z}_k = \begin{pmatrix} u_k \\ v_k \end{pmatrix}$ is the **state vector** at step k .

Long-term Behavior: Computing A^k

$$z_k = A^k z_0$$

Q1: Computing Numbers

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} 1.2 & -0.2 \\ 0.3 & 0.5 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix}$$

$$z_{t+1} = Az_t.$$

How to quickly calculate z_k ? Suppose $z_0 = [100, 40]^T$.

If A is diagonalizable, i.e. A can be written as $A = SDS^{-1}$, then $z_k = SD^k S^{-1} z_0$.
 $\downarrow$
 $A^k \cdot z_0$

Step 1: Compute S and D.

eigenvalues

	Value	Simple eigen-vector
λ_1	1.1	$x_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$
λ_2	0.6	$x_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$$\begin{cases} Ax_1 = \lambda_1 x_1 \\ Ax_2 = \lambda_2 x_2 \end{cases}$$

$$A \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \Rightarrow A = SDS^{-1}$$

$$S = [x_1 \ x_2] = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad D = \text{diag}(1.1, 0.6), \quad S^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

$$= \frac{1}{2 \times 3 - 1 \times 1} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

Remark:

Two eigenvalues are different, so A must be diagonalizable. You can keep this in mind; no need to say it explicitly.

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Formula for 2×2 matrix inverse:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad ad - bc \neq 0,$$

$$\Rightarrow A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Q1: Computing Numbers (cont'd)

Step 2: Compute $SD^k(S^{-1}v_0)$.

$$(c_1, c_2)^\top = S^{-1}z_0 = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 100 \\ 40 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 260 \\ -20 \end{pmatrix} = \begin{pmatrix} 52 \\ -4 \end{pmatrix}.$$

Let $\alpha_k := 1.1^k$, $\beta_k := 0.6^k$. Then $D^k = \text{diag}(\alpha_k, \beta_k)$.

$$\begin{aligned} z_k &= SD^k S^{-1} z_0 = \underbrace{SD^k}_{\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}} \underbrace{\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}}_{\begin{pmatrix} 52 \\ -4 \end{pmatrix}} \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1.1^k & 0 \\ 0 & 0.6^k \end{bmatrix} \begin{pmatrix} 52 \\ -4 \end{pmatrix} \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{pmatrix} 52 \cdot 1.1^k \\ -4 \cdot 0.6^k \end{pmatrix} \\ &= \begin{pmatrix} 104 \cdot 1.1^k - 4 \cdot 0.6^k \\ 52 \cdot 1.1^k - 12 \cdot 0.6^k \end{pmatrix} \end{aligned}$$

Note: When $k \gg 1$, $0.6^k \ll 1$,
(e.g. $0.6^k < 0.01$ when $k > 10$)

Comment: When k is large, $z_k \approx (104 \cdot 1.1^k, 52 \cdot 1.1^k)$.

Roughly 10% increase every year;

Matrix Dynamic Simulator

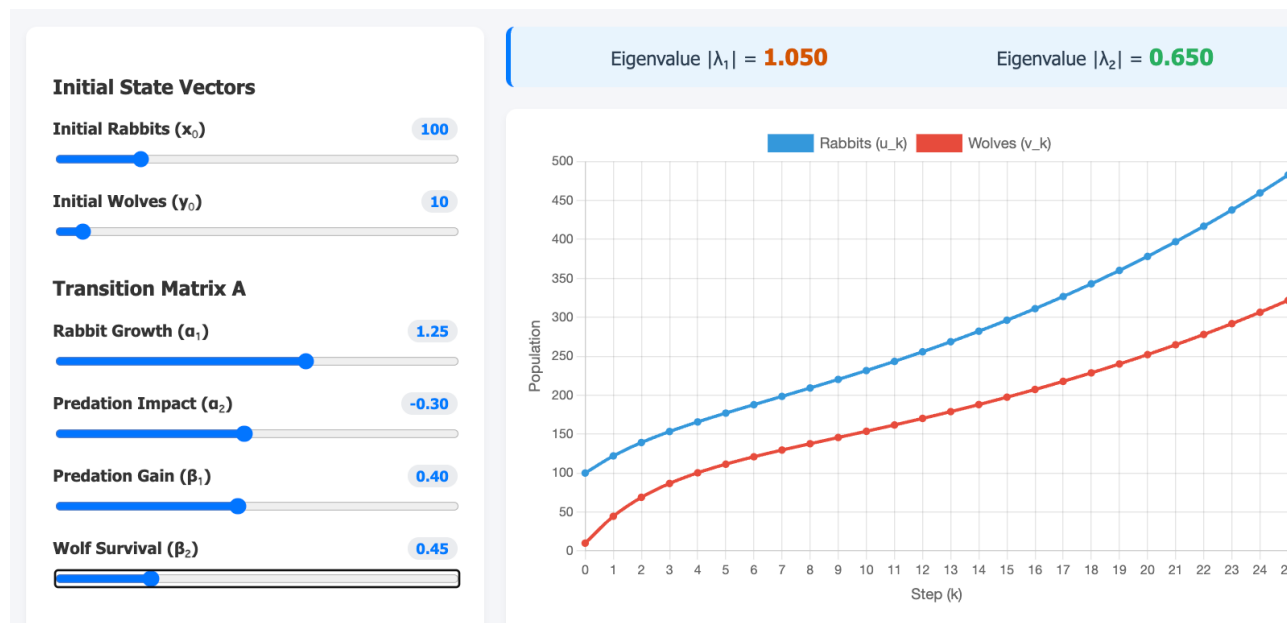
An **interactive tool** to show the dynamics (copy this link to your browser):
<https://codepen.io/sundirac/pen/pvEBeLX>

You can adjust the parameters to see different evolution dynamics.

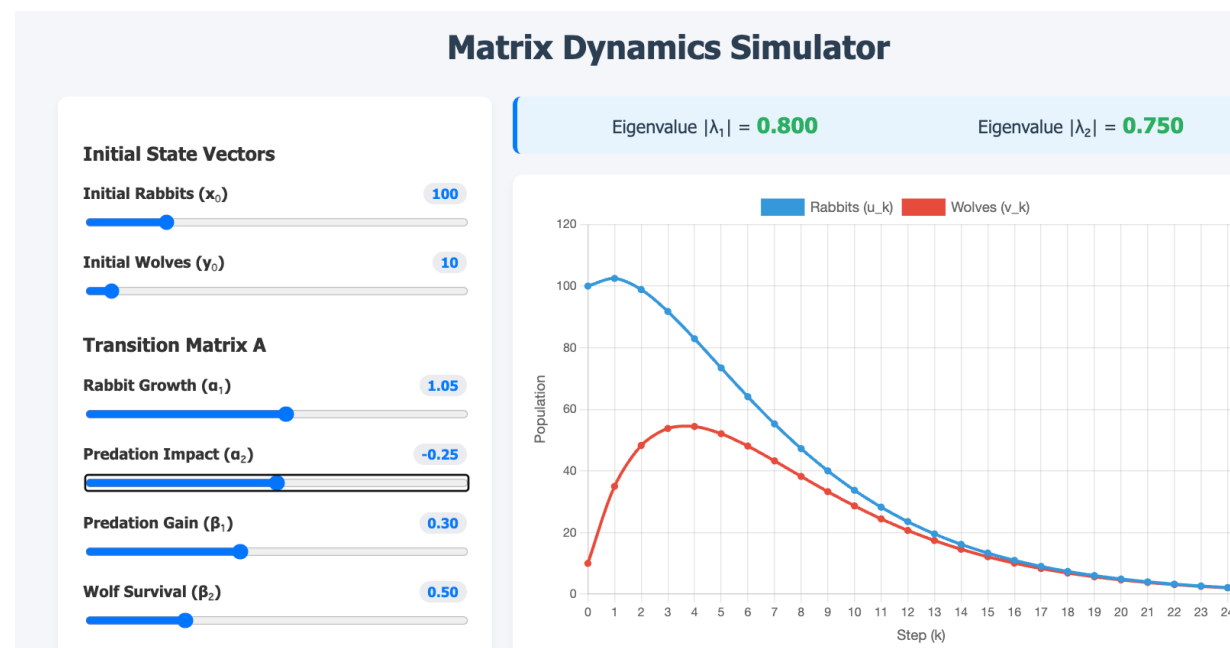


Playing with Simulator

You can play with the simulator.



Sometimes:
Both wolves & rabbits go up.



Sometimes:
Both wolves & rabbits diminish.

Question:
What determines the difference?

Q2 Converging to 0?

From $z_{t+1} = Az_t$, we have $z_k = A^k z_0$.

Then $\lim_{t \rightarrow \infty} z_t = \lim_{t \rightarrow \infty} A^t z_0 = \left(\lim_{t \rightarrow \infty} A^t \right) z_0$.

Claim 24.3: Suppose A is diagonalizable.

If all eigenvalues of A satisfy $|\lambda| < 1$, then $\lim_{t \rightarrow \infty} A^t = 0$.

Fact: If $\lambda \in \mathbb{C}$ and $|\lambda| < 1$, then $\lim_{t \rightarrow \infty} \lambda^t = 0$.

Proof of Claim 24.3: Suppose $A = SDS^{-1}$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$,

Then $\lim_{t \rightarrow \infty} D^t = \lim_{t \rightarrow \infty} \text{diag}(\lambda_1^t, \dots, \lambda_n^t) = \text{diag}(0, \dots, 0) = 0_{n \times n}$.

$$\lim_{t \rightarrow \infty} A^t = \lim_{t \rightarrow \infty} S \cdot D^t \cdot S^{-1} = S \cdot 0 \cdot S^{-1} = 0_{n \times n}.$$

Q2 Converging to 0?

e.g. $A = \begin{pmatrix} 1.2 & -0.6 \\ 0.4 & 0.1 \end{pmatrix}$, then : $\lambda_1 = 0.9$, $\lambda_2 = 0.4$

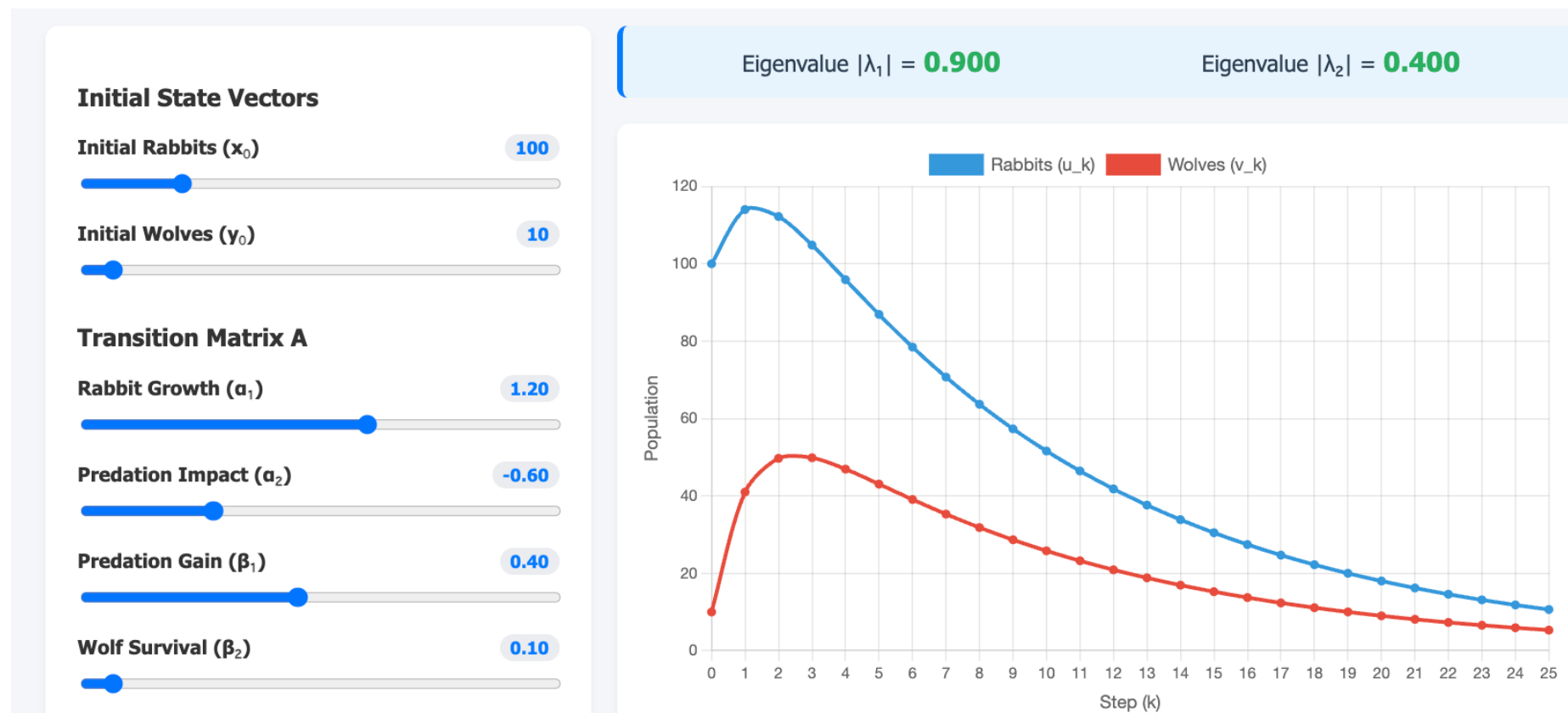
Because the eigenvalues are distinct, and satisfy $|\lambda_{1,2}| < 1$, the matrix is diagonalizable and:

$$A^t \xrightarrow{t \rightarrow \infty} 0, \quad z_t = A^t z_0 \xrightarrow{t \rightarrow \infty} 0$$

for every non-zero $z_0 \in \mathbb{R}^2$.

No need to
Compute
eigenvectors

In this setting, no matter what initial values, **wolves & rabbits finally disappear.**



Benefits of Using Diagonalization and Eigenvalues

In the example of “wolf-rabbit” dynamics, diagonalization and eigenvalues provide following benefits:

First, can provide a **precise formula** for the numbers at any time.
(simple to compute)

Second, reveal an interesting criterion **for judging “flourish of disappear”**:
modulus of eigenvalues all < 1 or some > 1 .

Third, this formula can provide an **estimate of increasing ratio** or decreasing ratio (when one eigenvalue’s modulus is much larger than another)



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:

Summary Today (Instructor)

One sentence summary:

We learned similar matrix, diagonalizable and computing dynamics.

Detailed summary:

i) Similar matrix: $A = SBS^{-1}$

Property: Similar matrices have the same multiset of eigenvalues.

ii) Diagonalizable: some matrix $A = SDS^{-1}$, where D is diagonal.

--Diagonalizable iff A 's eigenvectors can form a basis.

$$\text{iii)} \quad z_{k+1} = Az_k, \forall k \quad \Rightarrow \quad z_k = A^k z_0$$

$$\Rightarrow [\text{if } A = SDS^{-1}] \quad z_k = S D^k S^{-1} z_0.$$

If all eigenvalues of A satisfy $|\lambda| < 1$, then $z_k \rightarrow 0$, as $k \rightarrow \infty$.